where w_s^0 is the weight of the standard added, and R_s and R_{u+s} the radioactivities measured from the substoichiometrically separated samples of the standard and of the mixture of standard and unknown. The specific activities need not be determined as in (18.8) because equal weights of standard (w_s) and sample plus standard (w_{u+s}) are isolated by using the substoichiometric principle.

The usefulness of this technique for routine determination of numerous chemical elements in various matrixes has been demonstrated by Ruzicka, Stary, and others. It is also applicable to organic compounds and known in medicine as radioimmunoassay.

18.4. APPLICATIONS TO GENERAL CHEMISTRY

In no other area have radioactive tracers played such an important role as in the studies of chemical and biological reaction paths. This is, of course, due to that, in principle, in each radioactive decay the atom announces its position. Thus the detection sensitivity can approach the ultimate limit. A radioactive nuclide, as for example ^{14}C (β^- $t_{½}$ 5730 y), ^{32}P (β^- $t_{½}$ 14.282 d), or ^{198}Au ($\beta^-\gamma$, $t_{½}$ 2.6935 d), can be followed through a number of different chemical reaction steps, revealing details of metabolic or process reactions impossible to discover by other techniques.

18.4.1. Determination of chemical reaction paths

The use of radioisotopes in the study of the steps in a chemical reaction system is well established. Let us consider a few examples to illustrate this technique.

If phenol is labelled with deuterium or tritium in the hydroxyl group and warmed to a temperature slightly below decomposition, the labelled hydrogen migrates to other hydrogen positions of the benzene ring either by intra-molecular rearrangement or by inter-molecular reactions. However, using C_6H_5OT and C_6H_4TOH yields C_6H_4TOT, which can be formed only through inter-molecular collisions, thus eliminating intra-molecular rearrangement as the reaction mechanism.

The study of the reaction steps in the photosynthesis of carbohydrates from atmospheric CO_2 in the presence of light and chlorophyll is an outstanding example of the value of the tracer technique. The overall process (which involves many steps) can be written as

$$6\ CO_2 + 12\ H_2O \xrightarrow[\text{chlorophyll}]{\text{light}} C_6H_{12}O_6 + 6\ O_2 + 6\ H_2O \qquad (18.18)$$

Using partition chromatographic technique and tracers of ^{14}C, ^{32}P, and T, Calvin and co-workers were able to identify the intermediate steps involved. The experimental procedure is usually as follows (Fig. 18.7). Plants are placed in atmospheres containing ^{14}C-labelled CO_2 and irradiated with light. After different irradiation times the plants are removed and sections are digested to dissolve the material. A few drops of the solution

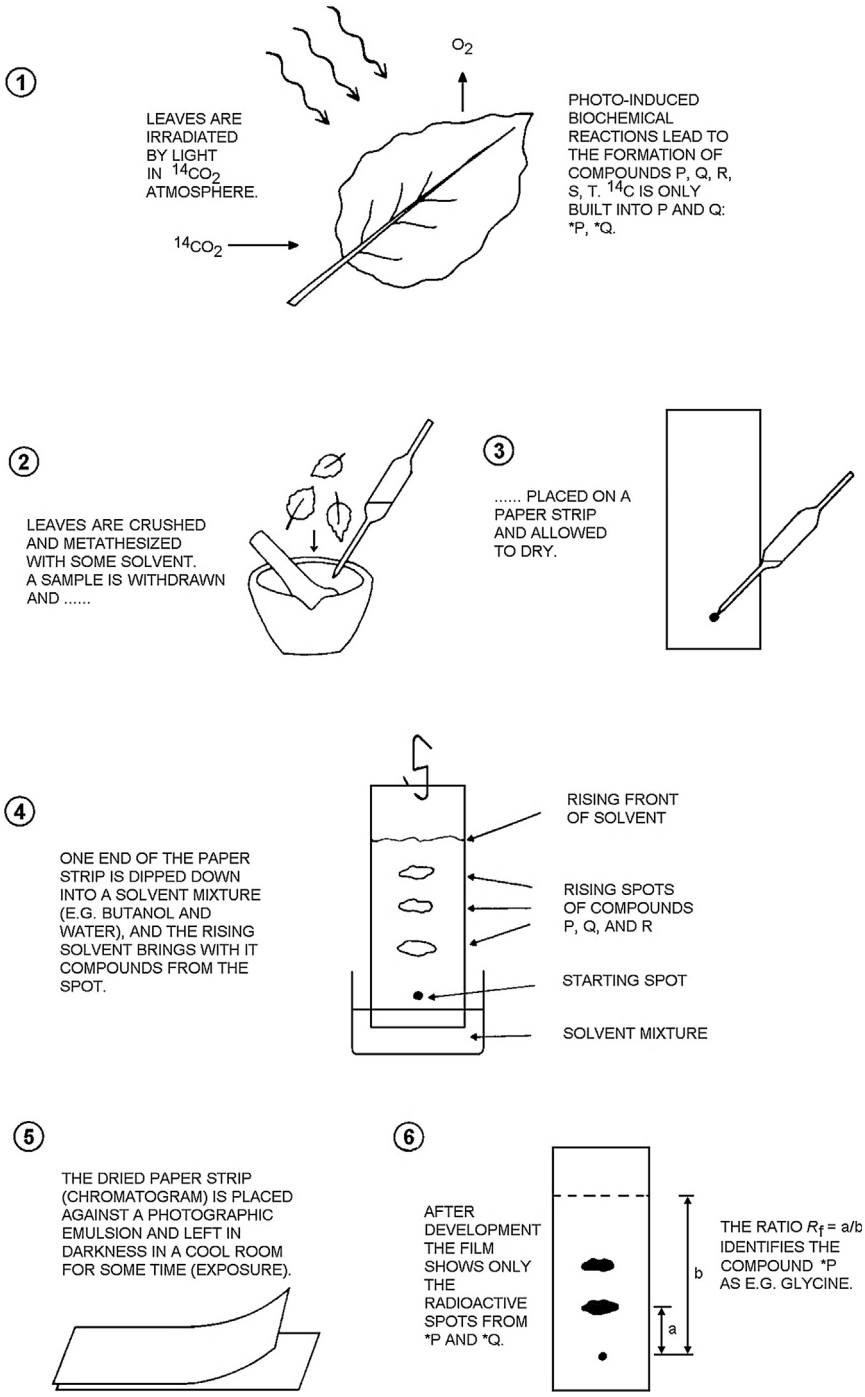


Figure 18.7 Paper chromatographic method for determination of which of components P, Q, R, S and T in a leaf are involved in photosynthesis.

containing the substance to be separated (metal ion, organic molecule, etc.) are placed a few centimeters from the end of a paper strip. The paper strip is hung vertically and dipped into a solution so that the initial point of placement of the substance is near the bottom of the strip above the solution level. Capillary forces draw the solution upwards and bring it into contact with the adsorbed substances at the starting point. As this occurs the substance moves a certain distance up the paper from the starting point with the distance traveled dependent on the kind of paper, the solution used, and the chemical properties of the substance. In such experiments, a certain R_f value for each substance can be defined as

$$R_f = \frac{\text{distance the substance of interest has travelled}}{\text{distance the liquid front has travelled}} \tag{18.19}$$

A typical solvent system for metal ions may be a mixture of acetone, dilute HCl, etc., while for organic substances it is possible to use mixtures of phenol and water, acetylacetonate and water, etc. Figure 18.8 shows a two-dimensional paper chromatogram; in this case it has been run initially with a particular solvent mixture, then (after drying) turned 90° and run with a second solvent mixture, thus increasing the selectivity of the separation. The separated substance can be quantitatively recovered by cutting out the spot and leaching the compound from it. The substance is identified either from its R_f value or further analyzed by standard methods.

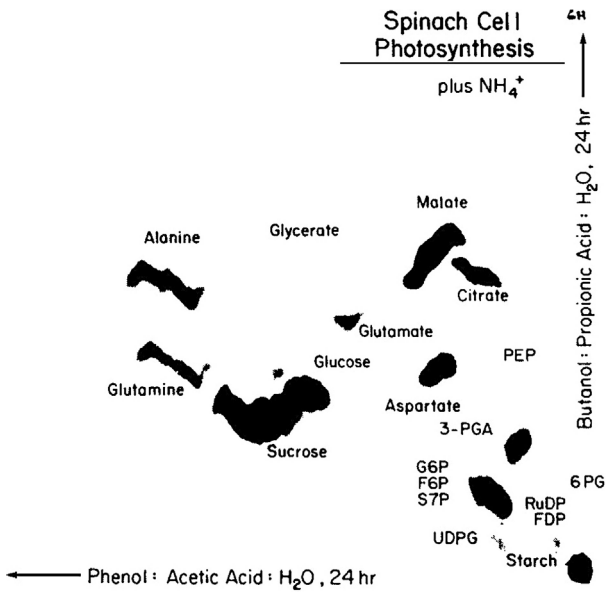


Figure 18.8 Two-dimensional paper radiochromatogram of ^{14}C-labeled products in photo-synthesis after exposure to $^{14}CO_2$ atmosphere. *(Courtesy Calvin and Lemmon.)*

18.4.2. Determination of chemical exchange rates

If two different chemical species with some element in common are mixed in solution, exchange of this common component may occur. The chemical equation would have the form

$$AX + BX^* \rightleftarrows AX^* + BX \tag{18.20}$$

Since the type and concentration of the chemical species remain unchanged, it is impossible to observe the exchange unless the atoms in one reactant are labelled. By using X^*, a radioactive isotope of X, the reaction may be followed, and at equilibrium the activity should be uniformly distributed between the two chemical species, i.e. the specific activity of X^* will be the same for both AX and BX. Of course, if AX and BX are both strong electrolytes, uniform distribution is essentially immediate upon mixing. If at least one of the reactants is an inorganic complex or an organic molecule, the exchange may be measurably slow if it occurs at all.

Since the chemical form of the reactants is not altered by the isotopic exchange, there is no change in heat content. However, the entropy of the total system is increased when uniformity in the distribution of the isotopes of X is achieved throughout the system. This entropy increase provides a decrease in the free energy, making isotopic exchange a spontaneous reaction. Despite this spontaneity, the exchange may be prevented or made very slow by a large energy of activation requirement in the formation of a necessary transition state.

For the exchange reaction represented above, the rate of increase of AX^* is equal to the rate of formation minus the rate of destruction of AX^*. The rate of formation is the product of the rate of reaction k_r, the fraction of reactions which occur with an active BX^*, and the fraction of reactions which occur with an inactive AX. Using the following notation

$$a = [AX] + [AX^*] \tag{18.21}$$

$$b = [BX] + [BX^*] \tag{18.22}$$

$$x = [AX^*] \tag{18.23}$$

$$y = [BX^*] \tag{18.24}$$

the rate of formation k_f is equal to

$$k_f = k_r(y/b)(a - x)/a \tag{18.25}$$

In a similar fashion, the rate of destruction k_d is equal to

$$k_d = k_r(x/a)(b - y)/b \tag{18.26}$$

Therefore

$$dx/dt = k_f - k_d = k_r(ay - bx)/(ab) \tag{18.27}$$

The solution of this equation is

$$\ln(1 - F) = -k_r t\,(a + b)/(ab) \tag{18.28}$$

where $F = x_t / x_\infty$ (x_∞ is the value of x_t at $t = \infty$, i.e. equilibrium). The rate of exchange k_r is evaluated from the slope of a plot log $(1 - F)$ versus t. If more than one rate of exchange is present due to exchange with nonequivalent atoms in a reactant, it may be difficult to resolve this curve sufficiently to obtain values for the reaction rates. Isotopic exchange is a standard tool of the scientist studying the kinetics of chemical reactions whose half-lives are longer than a minute.

One example of isotope exchange can be used to illustrate the value of these studies. Consider the exchange between di- and trivalent chromium in $HClO_4$ solutions. If the total chromium ion concentration is 0.1 M, it takes 14 days for the exchange to reach 50% completion at room temperature. In as much as the di- and trivalent cations are both positively charged, it is unlikely that they can approach each other closely enough to exchange an electron directly to allow a reversal of oxidation state, and a more likely mechanism is that an anion is involved as a bridge between the two cations such that the intrusion of the anion reduces the repulsion between the two cations. If this model of the isotopic exchange mechanism is valid, it could be proposed that the reaction mechanism would be

$$Cr(III)^* + X^- + Cr(II) \;\rightarrow\; [Cr^*{-}X{-}Cr]^{4+} \;\rightarrow\; Cr(II)^* + X^- + Cr(III) \tag{18.29}$$

Such a reaction mechanism would be fostered by the presence of anions that form complexes more readily than perchlorate ion. If HCl solutions are used rather than $HClO_4$ it is found that the exchange takes place more rapidly and the half time of exchange is only 2 min, which agrees with the proposed mechanism since chloride ions are known to be more favorable to complex formation than perchlorate ions. Without the use of radioactive (or isotope separated) chromium to label one of the original oxidation states there would be no means of identifying the exchange.

18.4.3. Determination of equilibrium constants

Determination of cation (M^{z+}) – anion (L^-, for ligand) solution equilibria can advantageously be done using radioactive metal tracers because of the ease with which the metal concentration can be measured, as e.g. in the determination of *solubility products*

$$K_{s0} = \left[M^{z+}\right]\left[L^-\right]^z \tag{18.30}$$

or *complex formation constants*, also named *stability constants*

$$\beta_n = \left[ML_n^{z-n}\right] / \left[M^{z+}\right]\left[L^-\right]^n \tag{18.31}$$

At trace concentrations of the metal (i.e. $[M]_{tot} \ll [L^-]$), the complexing anion concentration is unaffected by the metal ion concentration, which allows easy calculation of the values of these two important variables in the system. The experimental techniques used for this purpose are based on two-phase equilibria: solubility (solid/liquid), paper electrophoresis (paper/aqueous solution), solvent extraction (organic solvent/water) and ion exchange (resin/water). The equilibria measured have been shown to be independent of the metal concentration in the range 10^{-3} to $<10^{-13}$ M, as long as no polynuclear complexes (i.e. with several metal atoms per complex molecule) are formed (cf. §18.2.3).

(a) *Solubility*

The sensitivity of tracer detection makes measurement of solubilities relatively simple. This is illustrated by the first radioactive tracer experiment by de Hevesy and Paneth in 1913 in which the solubility of lead chromate was determined. Chromate ions were added to a solution of $PbCl_2$ containing a known amount of ^{210}Pb, precipitating the lead as $PbCrO_4$. The precipitate contained 2030 "radioactive units", and had a weight of 11.35 mg. The specific activity was thus $2030/11.35 = 179$ "units" mg^{-1}. Shaking the precipitate with water dissolved 2.14 units per 1000 ml. The solubility was calculated to be $2.14/179 = 0.012$ mg l^{-1} or 3.7×10^{-8} M Pb^{2+}. If $[Pb^{2+}] = [CrO_4^{2-}]$ the solubility product would be $K_{s0} = (3.7 \times 10^{-8})^2 = 1.4 \times 10^{-15}$. The modern value is 2×10^{-14}.

(b) *Solvent extraction*

As described in §18.2.6 the solvent extraction technique requires (i) a liquid two-phase system consisting of an organic solvent in contact with an aqueous solution, and (ii) the presence of an extractant (commonly a weak organic acid, abbreviated HA), which reacts with the metal ion to form an uncharged metal-organic complex MA_z, that preferentially dissolves in the organic phase. The distribution of the metal between the organic and aqueous phases can be shown to be fit the relation

$$D_M = K_{DC}\beta_z[A^-]^z \Sigma \beta_n [A^-]^n \tag{18.32}$$

where D_M is the distribution ratio of the metal as defined by eqn. (18.8), and K_{DC} is the *distribution constant* of the uncharged complex MA_z. $[A^-]$ is referred to as the free ligand ion concentration; using trace metal concentration, $[A^-]$ is easily calculated from the amount of acid, HA, added, its dissociation constant K_a, pH and the liquid volumes. From measurements of D_M as a function of $[A^-]$, the formation constants β_n for the complexes MA_n^{z-n} are calculated.

Figure 18.9 shows the distribution of lutetium (using trace concentrations of ^{177}Lu, ($\beta^-\gamma$ $t_{½}$ 6.71 d) between an aqueous solution and benzene containing an organic complex former (HA = acetylacetone). Eqn. (18.32) has been fitted to the experimental

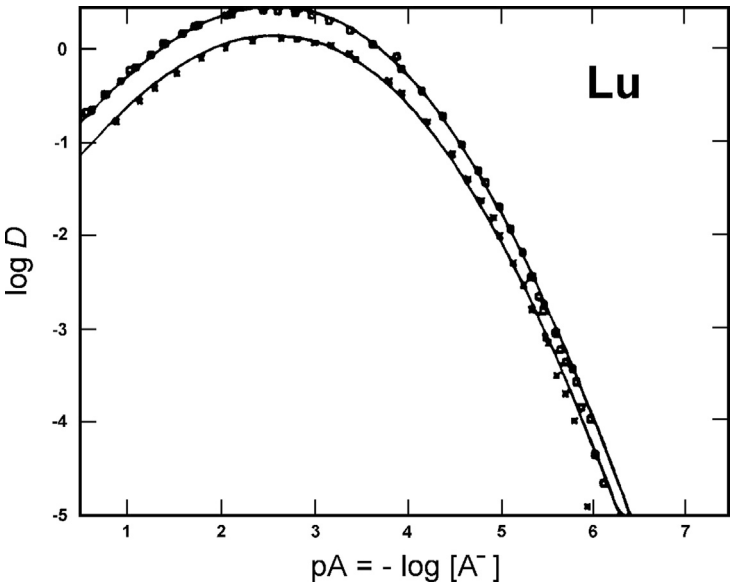


Figure 18.9 Distribution of ^{177}Lu (β^- $t_{½}$ 6.71 d) in 1 M $NaClO_4$/benzene, acetylacetone. Upper curve $[HAa]^0_{org}$ 3.0 M, lower curve 1.0 M. *(From Albinsson.)*

points by regression analysis to yield the equilibrium constants, by which the solid curves have been calculated.

Studies of this kind are often easy to do. However, the Lu-acetylacetone system was chosen as it was not as simple, yet good results were obtained by careful use of the tracer technique: Because the Lu concentration was 10^{-8} M, 10^{-5} M Nd^{3+} was added as *hold-back carrier* to avoid sorption losses (§18.2.3). At the pH's that had to be investigated Lu^{3+} (and also Nd^{3+}) may hydrolyse. Also, the very low *D*-values obtained (1 to 10^{-5}) are difficult to measure exactly with conventional technique. Therefore advanced techniques were developed, see Figure 18.10: (a), AKUFVE, which is a closed flow system, where the two liquid phases are mixed continuously and separated in a special flow-through centrifuge, and, after passing detection devices, returned to the mixing chamber. (b), LISOL, an "on-line" liquid-scintillation detection technique, in which a tiny fraction of the circulating phases is withdrawn, mixed (i) with acid to suppress sorption and hydrolysis, and (ii) with a liquid scintillator, and (iii) pumped to the PMT-detector. The LISOL avoids detector memory effects occurring in conventional on-line detectors and also allows measurements of pure α- and β-emitters (cf. §8.5.2).

As a second example we choose a chemically more complicated system, which, however, in practice is simpler than the one above, and therefore of more common use. The complexation of $^{237}Np^{4+}$ by HSO_4^- in 0.1 M $NaClO_4$ was studied by SX using the

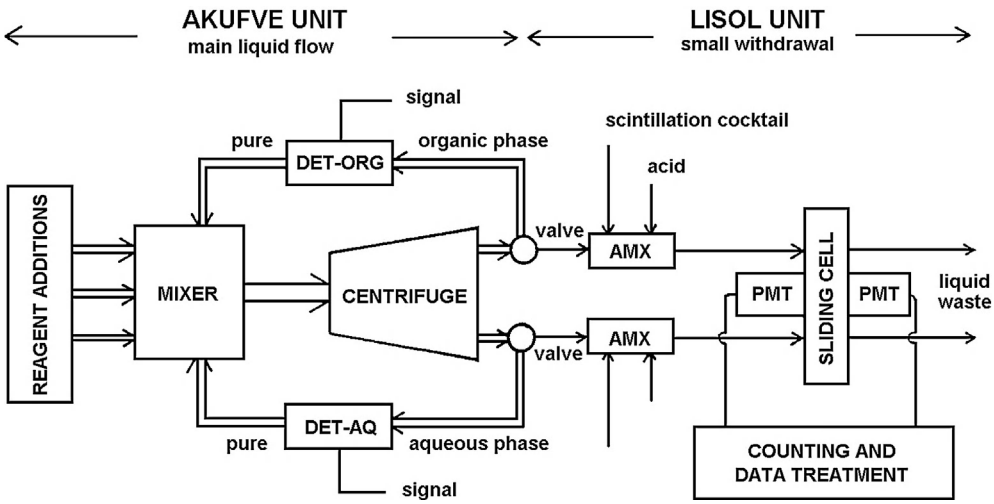


Figure 18.10 The AKUFVE-LISOL system used in investigating the system in Fig. 18.9.

system aqueous solution/$CHCl_3$ containing the organic extractant thenoyltrifluoroacetone (HTTA). In this case the relation

$$D_{\mathrm{Np}}/D_{\mathrm{p}} = \Sigma\beta_x \left[\mathrm{HSO}_4^-\right]^x \tag{18.33}$$

can be derived, where D_{Np} is the distribution ratio of Np in the presence, and D_{p} in the absence, of HSO_4^-; in this case pH and [HTTA] must be constant during the experiment, any change requiring correction. From the data in Figure 18.11 the formation constants β_x for Np(IV) sulphate complexes were determined.

The solvent extraction technique has been used extensively for studying complexation of metals for which it may not be possible or desirable to use macroscopic amounts, as e.g. for the heavy actinides or transactinides (Ch. 14).

(c) *Ion exchange*

The technique depends on the distribution of metal cations between a solid cation exchange resin and an aqueous solution. The exchange process for the M^{z+} ion takes place according to the equation

$$\mathrm{M}^{z+}(\mathrm{aq}) + z\mathrm{RH}(\mathrm{resin}) \;\rightleftarrows\; \mathrm{MR}_z(\mathrm{resin}) + z\mathrm{H}^+(\mathrm{aq}) \tag{18.34}$$

$$K_{\mathrm{iex}} = ([\mathrm{MR}_z][\mathrm{H}]^z)/([\mathrm{M}][\mathrm{RH}]^z) \tag{18.35}$$

omitting ionic charges and phase indices. Introducing D_{iex}, from (18.9), in (18.35) gives

$$D_{\mathrm{iex}} = K_{\mathrm{iex}}[\mathrm{RH}]^z\,[\mathrm{H}]^{-z} \tag{18.36}$$

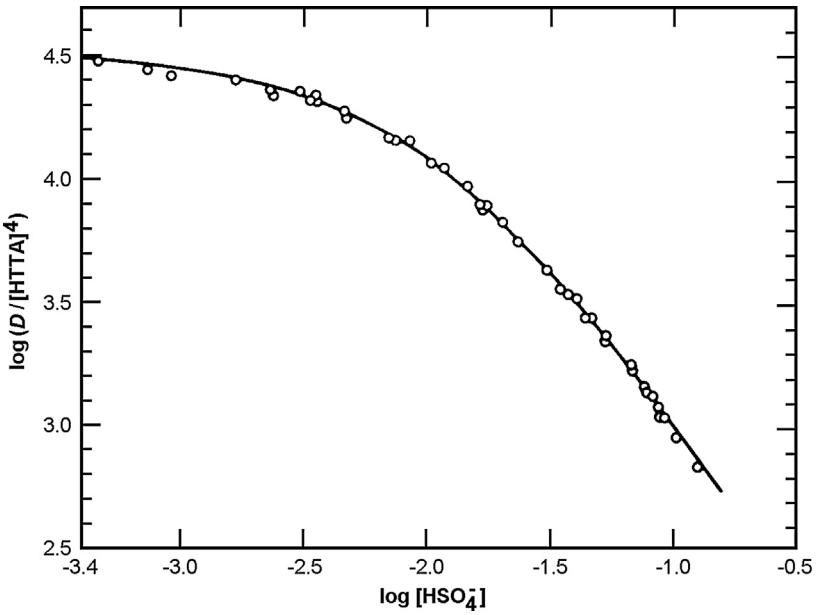


Figure 18.11 Normalized distribution ratio of ^{237}Np(IV) TTA complex between chloroform and an aqueous HSO_4^- solution. *(From Sullivan and Hindman.)*

showing that D_{iex} is a constant at constant solution/resin conditions (as in column ion exchange separations). However, when used for determining equilibrium constants in a system with several metal species in solution, each ionic species must be represented by an equation of type (18.35), leading to rather complicated expressions.

18.4.4. Studies of surfaces and reactions in solids

Surface properties of solids have been studied by dipping specimens into a solution containing a suitable radioactive tracer, and, after some "exposure time", removing them, washing their surface carefully and measuring the radiation emitted from them. It has been shown that a very rapid exchange takes place between atoms on a metal surface and the metal ions in solution. While the exchange is a function of the nature of the surface, within minutes it may involve atoms several hundred layers deep. The depth of penetration of sorbed radioactive isotopes can be obtained from a measurement of the absorption of the radiation or by measuring the radioactivity removed by cutting, or grinding, away thin layers. With the same technique the diffusion of atoms in their own solid matrix can be studied. For example, using single crystals of silver suspended in a solution containing silver nitrate labelled with ^{110m}Ag ($\beta^-\gamma$ $t_{½}$ 2418.76 d) it has been possible to demonstrate different rates of diffusion into different faces of the crystal. The surface area of solids can also be determined by

measurement of the sorption of radiotracers which do not penetrate into the specimen.

If a radioactive gas is incorporated in a crystalline compound the amount of gas released (the *emanation ability*) can be measured as a function of the temperature. It is found that the emanation increases considerably at certain temperatures, indicating structural changes in the solid at those temperatures. Studies of diffusion and emanation play a valuable role in understanding the mechanism of sintering and in the formation of new solid compounds. This has been of practical importance in the cement and glass industries, in the production of semiconductors, in the paint industry, etc. Studies of surface reactions are of practical importance for flotation, corrosion, metal plating and finishing, and detergent action to name only a few applications.

The distribution of a radioactive element or compound in a composite matrix can made visible either to the naked eye or under a microscope by means of *autoradiography*. The technique is based on the blackening of photographic films when exposed to nuclear radiation (cf. §§8.2 and 8.10). The technique is best illustrated by an example.

Lead is an unwanted impurity in stainless steel even in very small amounts. In order to investigate the mechanism of its incorporation, ^{212}Pb was added to a steel melt. After cooling, the ingot was cut by a saw and the flat surface machine-polished and etched in an electrolytic bath (electro-polishing). This provided a very flat and "virgin" surface. A photographic film was placed firmly with even pressure against the metal surface, and the film was exposed in darkness in a cool room for about a week. After development of the film, darkened spots caused by the radiation from ^{212}Pb showed where on the metal surface lead was present. By taking the results into account in the production process, the negative effect of lead in the raw material could be reduced.

In another technique, a polished surface of the specimen (metal, mineral, etc.) was dipped into a solution containing a radioactive reagent, which selectively reacts with one of the constituents of the surface. A mineral was dipped into potassium ethyl xanthate labelled with ^{35}S (β^- 87.5 d); the xanthate reacted selectively with sphalerite (zinc blende), ZnS, in the sample. The distribution of the xanthate, as shown by the autoradiograph, indicated the ZnS distribution in the mineral. The low β-energy of ^{35}S, E_{max} 0.2 MeV, was an advantage to the technique because the resolution of the autoradiograph increases with decreasing particle range.

18.5. APPLICATIONS TO LIFE SCIENCES

The largest field of application of radionuclides is in the life sciences. A survey is presented in Table 18.3.

In the reactions leading to the desired product (column A) many factors must be considered. They are of such importance that we devote several separate chapters to this: Ch. 10 on nuclear reactions, Ch. 16 on particle accelerators and Ch. 17 on production of